

Existence, Localization, and Stability of a Limit Cycle in a Class of Benchmark Biomolecular Oscillators: Supplementary Material

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The authors developed all algorithms presented in this supplementary. The LaTeX syntax and formatting were partially generated with the help of ChatGPT (OpenAI) as a support tool.

I. PART - I: COMPLETE TABLE III OF MAIN PAPER

Table I: Direction of flow of vector field within the smaller boxes.

Alt. Text: Complete table of Table III of the main paper. This shows the direction of flow between all the boxes after dividing the hypercube into 2^5 smaller hyperrectangles.

Box	Coordinates of the hyperrectangle	Direction of the flow of vector field
1	$0 \leq x_1 \leq x_0$ $0 \leq x_2 \leq x_0$ $0 \leq x_3 \leq x_0$ $0 \leq x_4 \leq x_0$ $0 \leq x_5 \leq x_0$	Box 2 $\uparrow$ Box 32 $\leftarrow$ Box 1 $\rightarrow$ Box 29 $\swarrow \quad \searrow$ Box 31 Box 32
2	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $0 \leq x_2 \leq x_0$ $0 \leq x_3 \leq x_0$ $0 \leq x_4 \leq x_0$ $0 \leq x_5 \leq x_0$	Box 25 $\nearrow$ Box 2 $\rightarrow$ Box 26 $\searrow$ Box 28
3	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $0 \leq x_3 \leq x_0$ $0 \leq x_4 \leq x_0$ $0 \leq x_5 \leq x_0$	Box 2 $\nearrow$ Box 3 $\rightarrow$ Box 18 $\searrow$ Box 27
4	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $0 \leq x_4 \leq x_0$ $0 \leq x_5 \leq x_0$	Box 10 $\nearrow$ Box 4 $\rightarrow$ Box 3 $\searrow$ Box 28
5	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $0 \leq x_5 \leq x_0$	Box 4 $\nearrow$ Box 5 $\rightarrow$ Box 17 $\searrow$ Box 19
6	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 5 $\uparrow$ Box 10 $\leftarrow$ Box 6 $\rightarrow$ Box 7 $\swarrow \quad \searrow$ Box 9 Box 8
7	$0 \leq x_1 \leq x_0$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 12 $\nearrow$ Box 7 $\rightarrow$ Box 13 $\searrow$ Box 14
Continued on the next page.		

Continuation of Table I ...		
Box	Coordinates of the hyperrectangle	Direction of the flow of vector field
8	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $0 \leq x_2 \leq x_0$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 11 ↗ Box 8 → Box 16 ↘ Box 17
9	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $0 \leq x_3 \leq x_0$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 12 ↗ Box 9 → Box 15 ↘ Box 19
10	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $0 \leq x_4 \leq x_0$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 13 ↗ Box 10 → Box 16 ↘ Box 18
11	$0 \leq x_1 \leq x_0$ $0 \leq x_2 \leq x_0$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 7 ↗ Box 11 → Box 21 ↘ Box 22
12	$0 \leq x_1 \leq x_0$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $0 \leq x_3 \leq x_0$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 12 → Box 24
13	$0 \leq x_1 \leq x_0$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $0 \leq x_4 \leq x_0$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 13 → Box 23
14	$0 \leq x_1 \leq x_0$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $0 \leq x_5 \leq x_0$	Box 5 ↗ Box 14 → Box 24 ↘ Box 27
15	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $0 \leq x_2 \leq x_0$ $0 \leq x_3 \leq x_0$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 8 ↗ Box 15 → Box 20 ↘ Box 26
Continued on the next page.		

Continuation of Table I ...		
Box	Coordinates of the hyperrectangle	Direction of the flow of vector field
16	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $0 \leq x_2 \leq x_0$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $0 \leq x_4 \leq x_0$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 16 $\rightarrow$ Box 21
17	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $0 \leq x_2 \leq x_0$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $0 \leq x_5 \leq x_0$	Box 17 $\rightarrow$ Box 28
18	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $0 \leq x_3 \leq x_0$ $0 \leq x_4 \leq x_0$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 9 $\nearrow$ Box 15 $\rightarrow$ Box 23 $\searrow$ Box 25
19	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $0 \leq x_3 \leq x_0$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $0 \leq x_5 \leq x_0$	Box 19 $\rightarrow$ Box 26
20	$0 \leq x_1 \leq x_0$ $0 \leq x_2 \leq x_0$ $0 \leq x_3 \leq x_0$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 11 $\nearrow$ Box 20 $\rightarrow$ Box 12 $\searrow$ Box 30
21	$0 \leq x_1 \leq x_0$ $0 \leq x_2 \leq x_0$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $0 \leq x_4 \leq x_0$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 21 $\rightarrow$ Box 13
22	$0 \leq x_1 \leq x_0$ $0 \leq x_2 \leq x_0$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $0 \leq x_5 \leq x_0$	Box 14 $\nearrow$ Box 22 $\rightarrow$ Box 17 $\searrow$ Box 31
23	$0 \leq x_1 \leq x_0$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $0 \leq x_3 \leq x_0$ $0 \leq x_4 \leq x_0$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 23 $\rightarrow$ Box 12
Continued on the next page.		

Continuation of Table I ...		
Box	Coordinates of the hyperrectangle	Direction of the flow of vector field
24	$0 \leq x_1 \leq x_0$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $0 \leq x_3 \leq x_0$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $0 \leq x_5 \leq x_0$	Box 24 $\rightarrow$ Box 19
25	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $0 \leq x_2 \leq x_0$ $0 \leq x_3 \leq x_0$ $0 \leq x_4 \leq x_0$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 15 $\nearrow$ Box 25 $\rightarrow$ Box 16 $\searrow$ Box 29
26	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $0 \leq x_2 \leq x_0$ $0 \leq x_3 \leq x_0$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $0 \leq x_5 \leq x_0$	Box 26 $\rightarrow$ Box 17
27	$0 \leq x_1 \leq x_0$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $0 \leq x_4 \leq x_0$ $0 \leq x_5 \leq x_0$	Box 4 $\nearrow$ Box 27 $\rightarrow$ Box 13 $\searrow$ Box 32
28	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $0 \leq x_2 \leq x_0$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $0 \leq x_4 \leq x_0$ $0 \leq x_5 \leq x_0$	Box 28 $\rightarrow$ Box 16
29	$0 \leq x_1 \leq x_0$ $0 \leq x_2 \leq x_0$ $0 \leq x_3 \leq x_0$ $0 \leq x_4 \leq x_0$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 20 $\nearrow$ Box 29 $\rightarrow$ Box 21 $\searrow$ Box 23
30	$0 \leq x_1 \leq x_0$ $0 \leq x_2 \leq x_0$ $0 \leq x_3 \leq x_0$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $0 \leq x_5 \leq x_0$	Box 22 $\nearrow$ Box 30 $\rightarrow$ Box 24 $\searrow$ Box 26
31	$0 \leq x_1 \leq x_0$ $0 \leq x_2 \leq x_0$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $0 \leq x_4 \leq x_0$ $0 \leq x_5 \leq x_0$	Box 21 $\nearrow$ Box 31 $\rightarrow$ Box 27 $\searrow$ Box 28
Continued on the next page.		

Continuation of Table I . . .		
Box	Coordinates of the hyperrectangle	Direction of the flow of vector field
32	$0 \leq x_1 \leq x_0$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $0 \leq x_3 \leq x_0$ $0 \leq x_4 \leq x_0$ $0 \leq x_5 \leq x_0$	Box 3 ↗ Box 32 → Box 23 ↘ Box 24

The direction of flow of vector field within the smaller hyperrectangles for a 5-gene cyclic regulatory network is given in the above table.

II. PART - II: INTERVAL-BASED REACHABILITY ANALYSIS

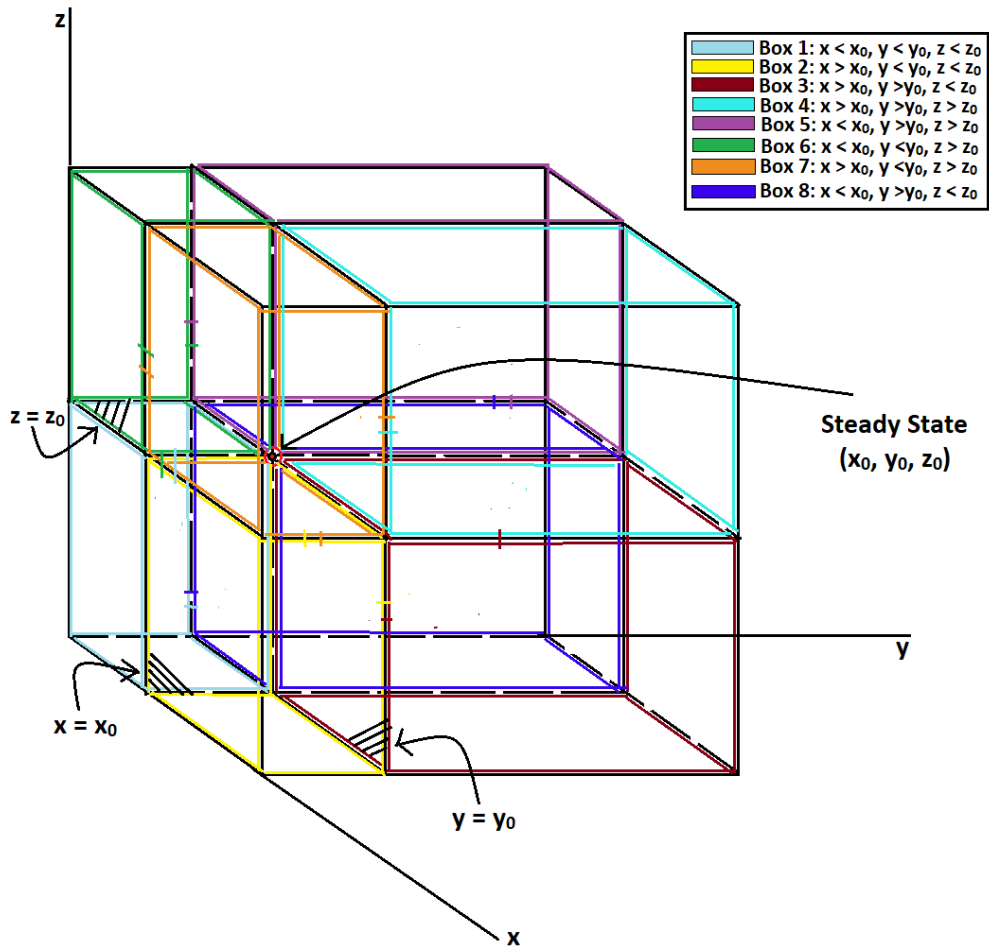


Figure 1: The cube is divided into 8 smaller boxes by the planes $x = x_0$, $y = y_0$ and $z = z_0$.

Alt. Text: The division of the invariant cube of the three-gene cyclic regulatory network, also called the repressilator, into 2^3 boxes.

Using the direction of the trajectories obtained from the signs of $\dot{x}, \dot{y}$ and $\dot{z}$, we get a continuous map of a cross-section onto itself. The map is as follows: $2 \rightarrow 7 \rightarrow 6 \rightarrow 5 \rightarrow 8 \rightarrow 3 \rightarrow 2$.

A. Problem-Statement

The objective is to extend the core idea of the Brouwer's Fixed Point theorem to an interval-based Reachability Analysis and localize the limit cycle trajectory. We aim to check whether a specified hyperrectangular region of initial conditions evolves in a manner such that the computed flowpipe either returns to and overlaps with or entirely traverses the initial region.

B. Methodology

The methodology involves dividing the search space into smaller hyperrectangles and evolving the system dynamics from each region using interval-based reachability analysis to generate a flowpipe representation. For every initial box X_0 , the computed flowpipe is examined to detect whether the reachable sets return back to (or intersect with) the initial region. Based on this return behavior, each region is classified as: (i) Does not contain the limit cycle, (ii) May contain the limit cycle, (iii) Contains the limit cycle, or (iv) Failed boxes.

This classification is performed using the return-index detection routine `intersection`, which identifies time indices where the reachable set overlaps with the initial box X_0 . If no such return indices exist, the region is classified as “Does not contain the limit cycle”. To evaluate whether the return behavior is consistent with a limit cycle being contained inside the region, a geometric containment check is applied on the reachable sets. Specifically, the initial reachable set is defined as

$$Z_0 = \text{set}(\mathcal{S}[1]),$$

and it is compared against all return candidates

$$Z_1 = \text{set}(\mathcal{S}[k]) \quad \forall k \in I,$$

where I is the return index set produced by the `intersection` function. A subset test `issubset`(Z_1, Z_0) is then used to determine whether the return set is enclosed within the initial reachable set approximation.

The final classification is performed as follows:

- **Does not contain the limit cycle** is assigned when no return indices are detected, meaning the flowpipe never intersects the initial region.
- **May contain the limit cycle** is assigned when return indices are detected, but none of the corresponding reachable sets satisfy the subset condition with respect to Z_0 , indicating that the trajectory returns to the initial region but the enclosure property fails.
- **Contains the limit cycle** is assigned when all return indices detected fulfill the subset condition `issubset`(Z_1, Z_0).
- **Failed boxes** are recorded when reachability computation fails due to solver breakdown or numerical over approximation, in which case no reliable classification can be made.

This approach avoids expensive direct geometric comparisons over the entire flowpipe and instead relies on return detection combined with subset verification, enabling limit cycle localization over a search space.

C. Algorithms

An external loop initially calls the function `divide` to partition the entire search space into smaller hyperrectangular subsets. Each subset is then analyzed sequentially, enabling classification of the phase space and localization of regions that may contain the limit cycle of the repressilator model. For each partition X_0 , the reachability flowpipe is computed using the subroutine `solve_ivp`, which propagates the interval initial condition set forward in time and generates an overapproximated reachable set sequence $\mathcal{S}$. If the solver fails or produces invalid bounds, the corresponding region is marked as a failed box and excluded from further analysis. If the flowpipe is successfully generated, the function `intersection` is used to identify the indices at which the reachable set intersects the original subset X_0 . These indices represent the return candidates. If no such return indices are obtained, the subset is classified as “Does not contain the limit cycle”.

Subsequently, a geometric containment verification is performed using a zonotope subset test. The initial reachable set $Z_0 = \text{set}(\mathcal{S}[1])$ is compared against each return candidate $Z_1 = \text{set}(\mathcal{S}[k])$, where k belongs to the intersection index set. The classification decision is made based on the subset condition `issubset`(Z_1, Z_0), which determines whether the return sets are enclosed within the initial approximation. This hierarchical structure allows categorization of the search space into regions (i) Does not contain the limit cycle, (ii) May contain the limit cycle, or (iii) Contains the limit cycle. The box containing the steady state is removed from the reachability analysis.

The algorithm outlined below defines the process for identifying and categorizing and thereby localizing limit cycle trajectory within the phase space of the repressilator.

Algorithm 1: Reachability Analysis for Localization of Limit Cycle

Input: Search space X_{search} , subdivisions n_{div} , skip parameter s
Output: $No_LC, LC_Partial, LC_Complete, Failed_Boxes$

▷ Initialize output lists.

 1 $No_LC \leftarrow [], LC_Partial \leftarrow [], LC_Complete \leftarrow [], Failed_Boxes \leftarrow []$

▷ Partition the search space.

 2 $S \leftarrow \text{divide}(X_{search}, n_{div})$

 3 **foreach** (X_0, idx) **in** S **do**

▷ Compute flowpipe using reachability analysis.

 4 $(\mathcal{S}, status) \leftarrow \text{solve_ivp}(X_0, idx)$

 5 **if** $status = false$ **then**

 6 Append idx to $Failed_Boxes$

 7 **continue**

 ▷ Check whether flowpipe returns to the initial box. Skip s used to avoid intersection of the initial time step sets (overapproximated zonotopes) with the initial box and also to speed up the iteration.

 8 $I \leftarrow \text{intersection}(\mathcal{S}, X_0, idx, s)$

 9 **if** $I = \emptyset$ **then**

 10 Append idx to No_LC

 11 **continue**

▷ Classify limit cycle containment

 12 $status_final \leftarrow \text{classify}(\mathcal{S}, X_0, I)$

 13 **if** $status_final = :complete$ **then**

 14 Append idx to $LC_Complete$

 15 **else if** $status_final = :partial$ **then**

 16 Append idx to $LC_Partial$

 17 **return** $(No_LC, LC_Partial, LC_Complete, Failed_Boxes)$

Algorithm 2: Subroutines for Reachability Analysis Based Localization of Limit Cycle

Input: Relevant function inputs

Output: Intermediate classification outputs

▷ **Subroutine: divide** (X_{search}, n_{div})

1 Partition X_{search} into smaller hyperrectangles

2 Assign indices to each partition

3 **return** list $S = \{(X_0, idx)\}$

▷ **Subroutine: solve_ivp** (X_0, idx)

4 Solve reachability IVP with TMJets21a

5 Compute flowpipe $\mathcal{S}$ as zonotope overapproximations

6 **if** *bounds contain Inf or solver crashes* **then**

7 | $status \leftarrow \text{false}$

8 **else**

9 | $status \leftarrow \text{true}$

10 **return** ($\mathcal{S}, status$)

▷ **Subroutine: intersection** ($\mathcal{S}, X_0, idx, s$)

11 Extract bounds of X_0 : $lb = \text{low}(X_0)$, $ub = \text{high}(X_0)$

12 $intersect_idx \leftarrow []$

13 **for** $i = s$ **to** $\text{length}(\mathcal{S})$ **do**

14 | $Z \leftarrow \text{set}(\mathcal{S}[i])$

15 | **if** Z intersects X_0 **then**

16 | Append i to $intersect_idx$

17 **return** $intersect_idx$

▷ **Subroutine: classify** ($\mathcal{S}, X_0, I$)

18 $Z_0 \leftarrow \text{set}(\mathcal{S}[1])$

19 $complete \leftarrow 0$, $partial \leftarrow 0$

20 **foreach** $k \in I$ **do**

21 | $Z_1 \leftarrow \text{set}(\mathcal{S}[k])$

22 | **if** $Z_1 \subseteq Z_0$ **then**

23 | $complete \leftarrow complete + 1$

24 | **else**

25 | $partial \leftarrow partial + 1$

26 **if** $complete == \text{length}(I)$ **then**

27 | **return** :complete

28 **else**

29 | **return** :partial

D. Results and Discussions

The above interval-based Reachability Analysis is used to classify the cross-section between boxes 2 and 7. This framework used here successfully divides the above cross-section of the repressilator according to their underlying dynamical behavior. The methodology, based on partitioning the phase space and analyzing the flowpipe of each partition, yielded various findings, which are visually summarized in Figure 2.

The analysis classified the vast majority of the partitioned region of this cross-section as not containing a periodic orbit. The black box in Figure 2, contains the unstable steady state (x_0, x_0, x_0) (the value of x_0 for this simulation is 5.361) of the system and is excluded from the Reachability Analysis. A distinct cluster of initial sets, marked in yellow in Figure 2, was classified as “May contain the limit cycle”. A cluster of green boxes around the box containing the steady state is classified as “Failed boxes”. This is because of the failure in the reachability analysis to return a solution. This can be attributed to solver crash or blow up in the bounds due to overapproximation. Further in Figure 3 the numerical solution of the repressilator model is plotted along with the surface between boxes 2 and 7. Here the trajectory passes through the yellow boxes in Figure 2, which was classified by the algorithm as “May contain the limit cycle”. This finding validates the method’s efficacy in localizing the region in which the limit cycle exists.

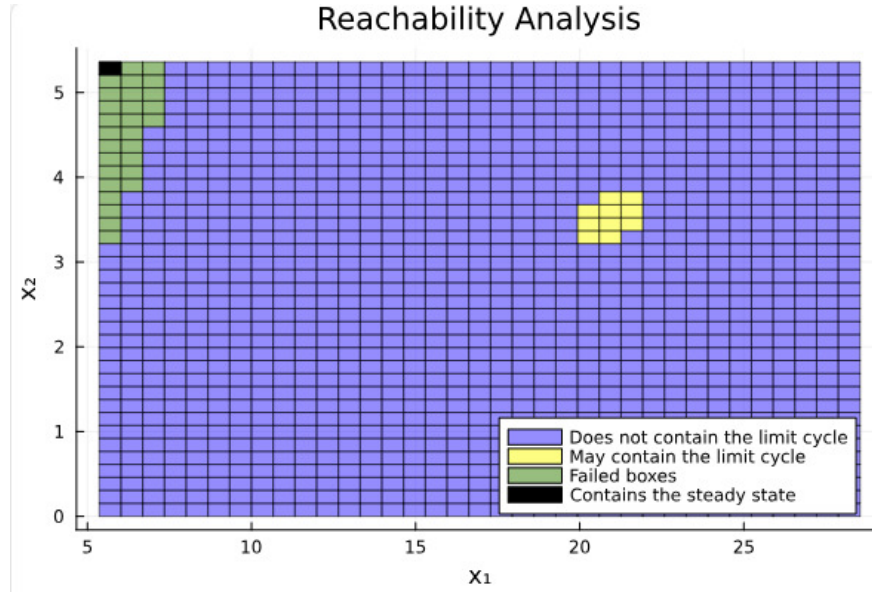


Figure 2: Projection of the surface between boxes 2 and 7 showing the geometric classification results. The black box contains the steady-state point and was excluded from the reachability analysis. The blue boxes is the region that does not contain the limit cycle. The cluster of yellow boxes signify the region that may contain the limit cycle. The green boxes are classified as the boxes that failed to return a solution for reachability analysis due to solver crash or blow up in the bounds due to overapproximation.

Alt. Text: Classification of the cross-section between box 2 and 7 of the repressilator using the proposed interval-based Reachability Analysis.

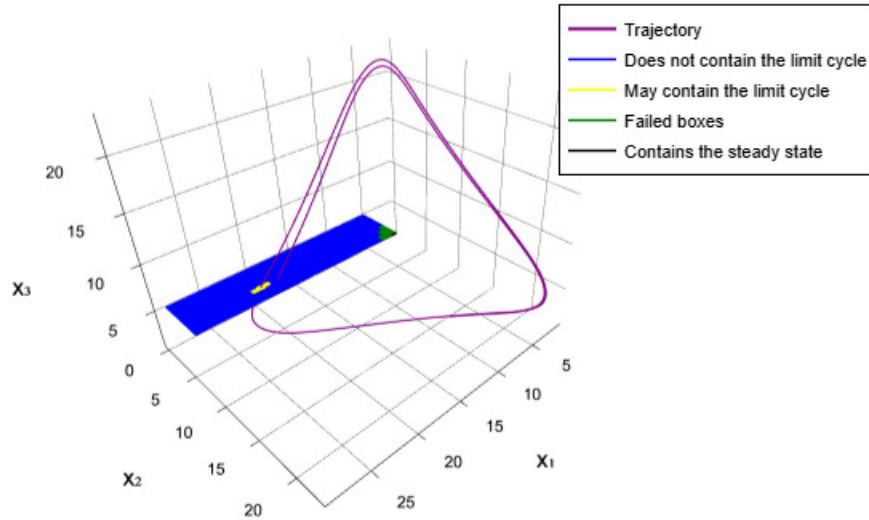


Figure 3: 3D plot of the numerical solution of the repressilator model. The resulting trajectory passes through the boxes marked as “May contain the limit cycle” in Figure 2.

Alt. Text: The numerical solution of the repressilator plotted along with the classification from Reachability Analysis of the cross-section between Box 2 and 7. The solution passes through the region classified as “May contain the limit cycle”.

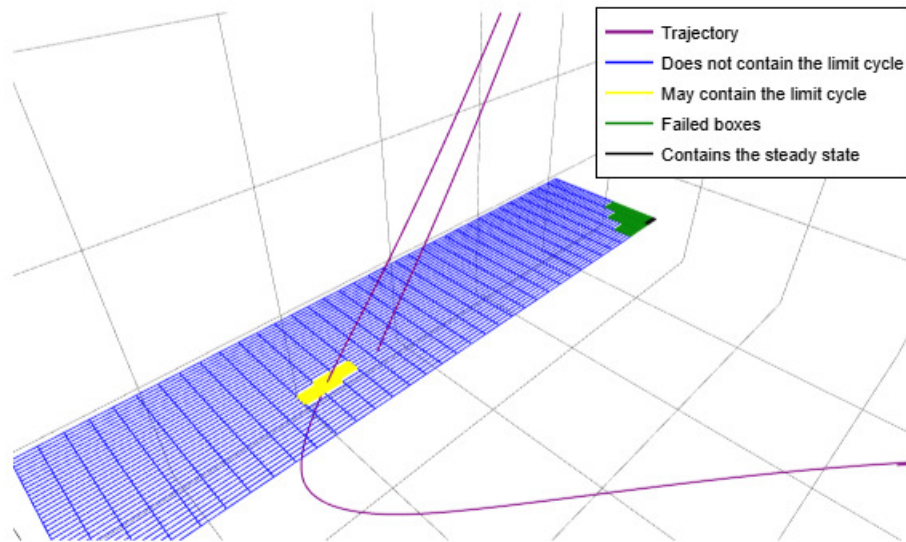


Figure 4: A magnified view of Fig. 2.

Alt. Text: A magnified view of Figure 3, giving a clear look at the numerical solution passing through the classified region.

E. Conclusion

In summary, the rigorous localization of the region of existence of the limit cycle in the repressilator relies on an interval-based Reachability Analysis, which, in conjunction with Brouwer's Fixed Point theorem, can provide a significantly tighter enclosure of the limit cycle trajectory.